# EXPERIMENT 5

## FAMILIARIZATION OF VARIOUS TOOLS OF THE SLEUTH KIT

**Date: 11/02/2026**

**AIM**

To familiarize various tools of sleuth kit

**THEORY**

The original part of Sleuth Kit is a C library and collection of command line file and volume system forensic analysis tools. The file system tools allow you to examine file systems of a suspect computer in a non-intrusive fashion. Because the tools do not rely on the operating system to process the file systems, deleted and hidden content is shown. It runs on Windows and Unix platforms.

The volume system (media management) tools allow you to examine the layout of disks and other media. The Sleuth Kit supports DOS partitions, BSD partitions (disk labels), Mac partitions, Sun slices (Volume Table of Contents), and GPT disks. With these tools, you can identify where partitions are located and extract them so that they can be analyzed with file system analysis tools.

When performing a complete analysis of a system, we all know that command line tools can become tedious. Autopsy is a graphical interface to the tools in The Sleuth Kit, which allows you to more easily conduct an investigation. Autopsy provides case management, image integrity, keyword searching, and other automated operations.

### Input Data

- Analyzes raw (i.e. `dd`), Expert Witness (i.e. E01/EnCase), VHD, VMDK, and AFF file system and disk images.
- Supports the NTFS, FAT, ExFAT, APFS, UFS 1, UFS 2, EXT2FS, EXT3FS, Ext4, HFS, ISO 9660, and YAFFS2 file systems (even when the host operating system does not or has a different endian ordering).

- Tools can be run on a live Windows or UNIX system during Incident Response. These tools will show files that have been "hidden" by rootkits and will not modify the A-Time of files that are viewed

**ALGORITHM**

1.Open command prompt

2.Type the following commands:

2.1.1mmls – Displays partition layout of a disk

2.1.2 fsstat – Shows file system details

2.1.3 fls – Lists files and directories in a forensic image

2.1.4 icat – Extracts file contents based on inode number

2.1.5 ifind – Finds the inode number of a file

**1. Analyzing Disk Images**

- `mmls` – Displays partition layout of a disk.

sh Copy Edit

```
mmls disk.img
```

- `fsstat` – Shows file system details.

sh Copy Edit

```
fsstat disk.img
```

**2. Listing Files & Directories**

- `fls` – Lists files and directories in a forensic image.

sh Copy Edit

```
fls -r -o 2048 disk.img
```

*(Here, `-r` recovers deleted files, `-o` specifies the partition offset.)*

- `ifind` – Finds the inode number of a file.

sh Copy Edit

```sh
ifind -n "filename.txt" disk.img
```

### 4. Extracting Metadata

- `istat` – Displays file metadata based on inode.

sh Copy Edit

```sh
istat disk.img 12345
```

### 5. Searching for Specific Data

- `grep` with TSK tools – Searching within file contents.

sh Copy Edit

```sh
strings disk.img | grep "password"
```

## OUTPUT

```
C:\Users\lab8-26\Downloads\s6\s6\sleuthkit-4.12.1-win32\bin>mmls C:\Users\lab8-26\Downloads/FTK.001
GUID Partition Table (EFI)
Offset Sector: 0
Units are in 512-byte sectors

      Slot      Start        End          Length       Description
000:  Meta      0000000000   0000000000   0000000001   Safety Table
001:  -------   0000000000   0000002047   0000002048   Unallocated
002:  Meta      0000000001   0000000001   0000000001   GPT Header
003:  Meta      0000000002   0000000033   0000000032   Partition Table
004:  000       0000002048   0000206847   0000204800   EFI system partition
005:  001       0000206848   0000239615   0000032768   Microsoft reserved partition
006:  002       0000239616   0445945855   0445706240   Basic data partition
007:  003       0445945856   0975220735   0529274880
008:  004       0975220736   0976771071   0001550336
```

```
C:\Users\lab8-26\Downloads\s6\s6\sleuthkit-4.12.1-win32\bin>fls -r -o 2048 C:\Users\lab8-26\Downloads\FTK.001
d/d 3:  EFI
+ d/d 38:       Microsoft
++ d/d 70:      Boot
+++ r/r 101:    BCD
+++ r/r 102:    BCD.LOG
+++ r/r 104:    BCD.LOG1
+++ r/r 106:    BCD.LOG2
+++ d/d 108:    bg-BG
++++ r/r 1287:  bootmgfw.efi.mui
++++ r/r 1290:  bootmgr.efi.mui
++++ r/r * 1293:        bootmgr.efi.mui.{c9c35566-
++++ r/r * 1296:        bootmgfw.efi.mui.{80f8093f
++++ r/r * 1299:        bootmgfw.efi.mui
++++ r/r * 1305:        bootmgr.efi.mui.{02736172-8c56-4f16-8690-09f70b3413de}
++++ r/r * 1308:        bootmgr.efi.mui
++++ r/r * 1314:        bootmgfw.efi.mui.{4853766a-f1d1-43a5-80df-ef4a228645f4}
+++ r/r 111:    kd_02_10df.dll
+++ d/d 113:    CIPolicies
++++ d/d 111846:        Active
+++++ r/r 111881:       {82443e1e-8a39-4b4a-96a8-f40ddc00b9f3}.cip
+++++ r/r 111886:       {CDD5CB55-DB68-4D71-AA38-3DF2B6473A52}.cip
+++++ r/r 111891:       {5DAC656C-21AD-4A02-AB49-649917162E70}.cip
+++++ r/r * 111899:     {CDD5CB55-DB68-4D71-AA38-3DF2B6473A52}.cip.{b07d0fe9-24d0-4eb7-bf4e-357f811ae042}
+++++ r/r * 111904:     {5DAC656C-21AD-4A02-AB49-649917162E70}.cip
+++ d/d 115:    cs-CZ
++++ r/r 113607:        bootmgfw.efi.mui
++++ r/r 113610:        bootmgr.efi.mui
++++ r/r 113613:        memtest.efi.mui
```

Activate Windows
Go to Settings to activate W

```
+++ r/r 474182: BCD.LOG
+++ r/r 474184: BCD.LOG1
+++ r/r 474186: BCD.LOG2
+ d/d 40:       Boot
++ r/r 421381:  bootx64.efi
++ r/r 421383:  fbx64.efi
++ r/r 421385:  mmx64.efi
+ d/d 42:       ubuntu
++ r/r 911974:  grubx64.efi
++ r/r 911976:  shimx64.efi
++ r/r 911978:  mmx64.efi
++ r/r 911979:  BOOTX64.CSV
++ r/r 911981:  grub.cfg
d/d 6:  System Volume Information
+ d/d * 911655: FveDecryptedVolumeFolder
+ d/d * 911656: _LIENT~1
+ d/d * 911659: AadRecoveryPasswordDelete
+ d/d * 911662: FveDecryptedVolumeFolder
+ d/d * 911666: ClientRecoveryPasswordRotation
+ d/d * 911669: AadRecoveryPasswordDelete
+ d/d * 911672: FveDecryptedVolumeFolder
+ d/d * 911676: ClientRecoveryPasswordRotation
+ d/d * 911679: AadRecoveryPasswordDelete
+ d/d * 911682: FveDecryptedVolumeFolder
v/v 3145731:    $MBR
v/v 3145732:    $FAT1
v/v 3145733:    $FAT2
V/V 3145734:    $OrphanFiles
```

Activate Windows

**RESULT**

Familiarized various tools of sleuth kits